

Demystifying Bayesian Model Fitting

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Scientific and statistical inference with Bayesian models

When you specify a model or hypothesis, you are essentially making a claim about the small world of your experiment and the data generated by it. The *model formula* specifies the relationship you claim between the variables of interest in this world. And the *prior* specifies your current state of knowledge (or belief) about those relationships. Bayesian model fitting then tells you what the data implies *within* this small world; whether that tells you anything about the real world is a scientific question, not a statistical one.

Consider the experimental world in which you hypothesize an effect of hours of sleep on a night on perceptual sensitivity the following day as a linear model. The model formula that describes it is:

$$d' = \beta_0 + \beta_1 \times \text{hours_sleep} + \mathcal{N}(0, \sigma)$$

Note that by choosing to model it as a linear model, you get with it all the assumptions made by linear models (normally distributed residuals, homoscedasticity, linear effects, etc).

The **priors** describe your current beliefs or informational state. This could be well informed or weakly informed (Figure 1). The priors essentially quantify your uncertainty about the parameters in the experimental world (in the above example, β_0 , β_1 , σ).

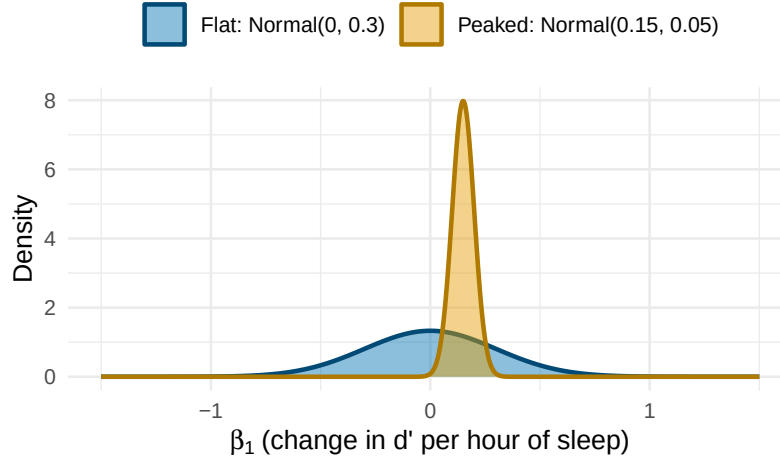


Figure 1: Two example priors for the effect of sleep on perceptual sensitivity (β_1). The peaked prior, $\text{Normal}(0.15, 0.05)$, has most of its probability in a narrow range of values: it expresses a confident, well-informed belief (lower uncertainty). The flat prior, $\text{Normal}(0, 0.3)$, spreads its probability over a much wider range: it expresses a weakly informed belief (higher uncertainty). The width (standard deviation) of the prior is what encodes this uncertainty: the wider the distribution, the less you claim to know about β_1 before seeing the data. The position (mean) of a prior encodes which value you consider most plausible.

Priors are typically informed by previous research, or can be reasonable guesses based on what you already know about your measures. For example (see Figure 2):

- **Perceptual sensitivity** (d') in tasks calibrated near threshold usually lies between about 0.5 and 2.5; values above 4 means performance is essentially at ceiling. This gives a sensible range for the intercept β_0 , and we may specify a prior like $\text{Normal}(1.5, 1)$.
- **Effects** should be scaled to the measure. If d' is around 0.5–2.5, one extra hour of sleep is unlikely to shift it by a full unit; changes of a few tenths are far more plausible. A prior like $\text{Normal}(0, 0.5)$ on β_1 allows for effects of this size in either direction.
- **Residual variability** (σ) must be positive, and should be on the order of how much d' varies across blocks (a few tenths), not several units. A distribution that is only defined for positive values, like a Half-Normal(0, 0.5), would be a good choice.
- **Another example:** reaction times in simple perceptual decision tasks typically fall between about 200 and 1000 ms. If you were modelling reaction times, a prior on the intercept that puts substantial probability below 100 ms or above 3 s is claiming something you already know to be implausible.

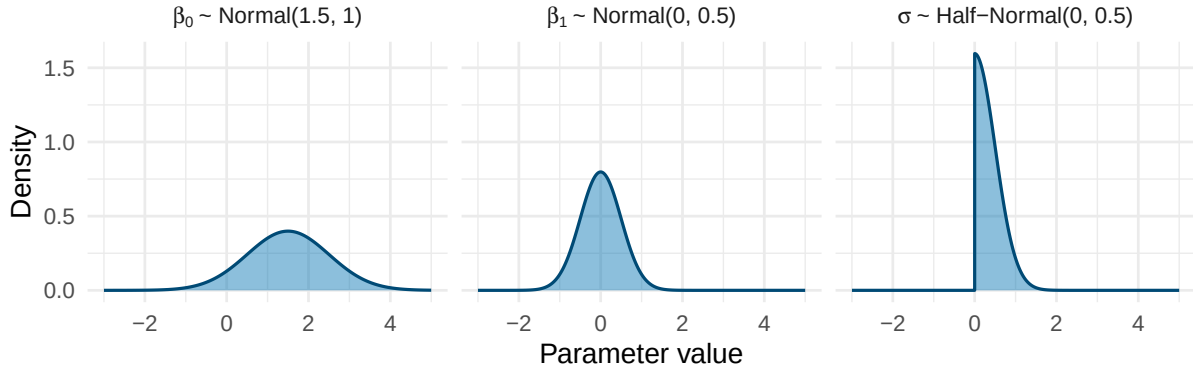


Figure 2: The priors for the sleep model’s three parameters. Left: the intercept β_0 , $\text{Normal}(1.5, 1)$. Middle: the sleep effect β_1 , $\text{Normal}(0, 0.5)$. Right: the residual standard deviation σ , $\text{Half-Normal}(0, 0.5)$, which is zero below 0. See last paragraph for details.

While Bayesian model fitting packages may provide reasonable defaults, it is always a good exercise to specify your priors yourself. To understand why, it helps to distinguish two kinds of inference. *Statistical inference* happens inside the model: given the model formula, the priors and the data, Bayes’ theorem mechanically turns the prior into the posterior. This is purely numerical inference. *Scientific inference* is everything around it: choosing a model formula and priors that reflect your theory and what you already know, and interpreting what the resulting posterior means for that theory. Your model formula and priors are how your scientific knowledge enters the statistical machinery, but scientific inference goes further than that: “the *meaning* of statistical analysis is not found in the data [or the model], but rather in our theories” - McElreath.

This document currently does not cover prior selection more broadly (remember weakly regularizing priors and sensitivity analysis; maybe in the future).

From prior to posterior

Bayesian model fitting is the process of updating the prior (the distribution of parameter values that you thought plausible before seeing the data) in light of the data, using Bayes’ theorem, to obtain the *posterior* (the distribution of parameter values that remain plausible after seeing the data).

Conceptually, this works as follows. To keep things simple and easy to visualize, we will only consider the two coefficients (β_0 , β_1) and ignore the residual standard deviation (σ) for now. Imagine drawing samples from your priors: each time, you draw one value from the prior for β_0 and one from the prior for β_1 . Each draw is therefore a pair (β_0 , β_1), and each pair describes one hypothetical experimental world. If you repeat this many times, the pairs build up the *joint prior distribution*: how plausible you consider each combination of intercept and slope before seeing the data (Figure 3).

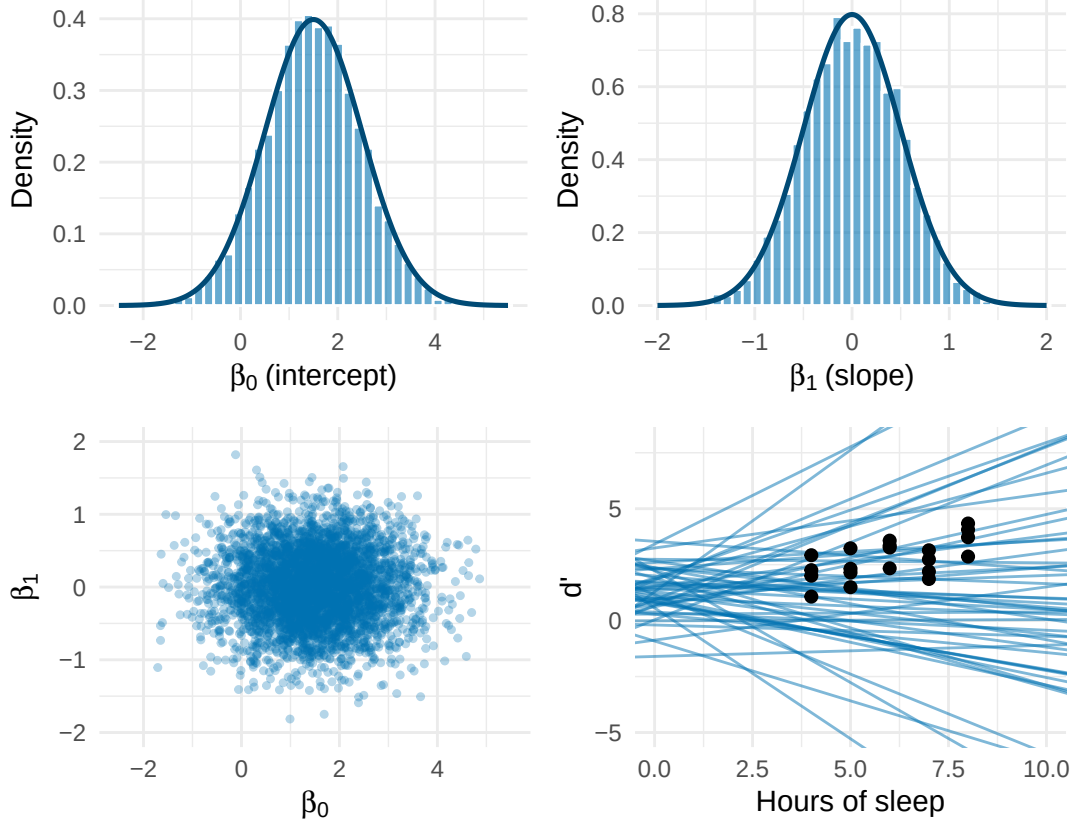


Figure 3: Samples from the priors (4000 draws). Top: histograms of the β_0 and β_1 draws, with the exact prior curves overlaid. Bottom left: the joint prior; each point is one (β_0, β_1) draw, so denser regions are combinations drawn more often. Bottom right: 50 of the draws, each shown as the line it implies between hours of sleep and d' . Black points are your experimental data. Many of the prior lines lie far from the data, or predict large negative values, showing that some of your prior beliefs represent very unrealistic predictions of the experimental world.

Each prior draw is a candidate world: a specific line relating hours of sleep to d' . For each candidate, we can ask: *if this were the experimental world, how probable is the data we actually observed?* This quantity is the **likelihood**. Lines that pass close to the data points make the observed data quite probable; lines that miss them make it very improbable.

Bayes' theorem combines the two sources of information:

$$\underbrace{p(\beta_0, \beta_1 \mid \text{data})}_{\text{posterior}} \propto \underbrace{p(\text{data} \mid \beta_0, \beta_1)}_{\text{likelihood}} \times \underbrace{p(\beta_0, \beta_1)}_{\text{prior}}$$

In other words: how plausible a candidate world is *after* seeing the data is proportional to how plausible the candidate world was *before* (the prior), weighted by how well it predicts the

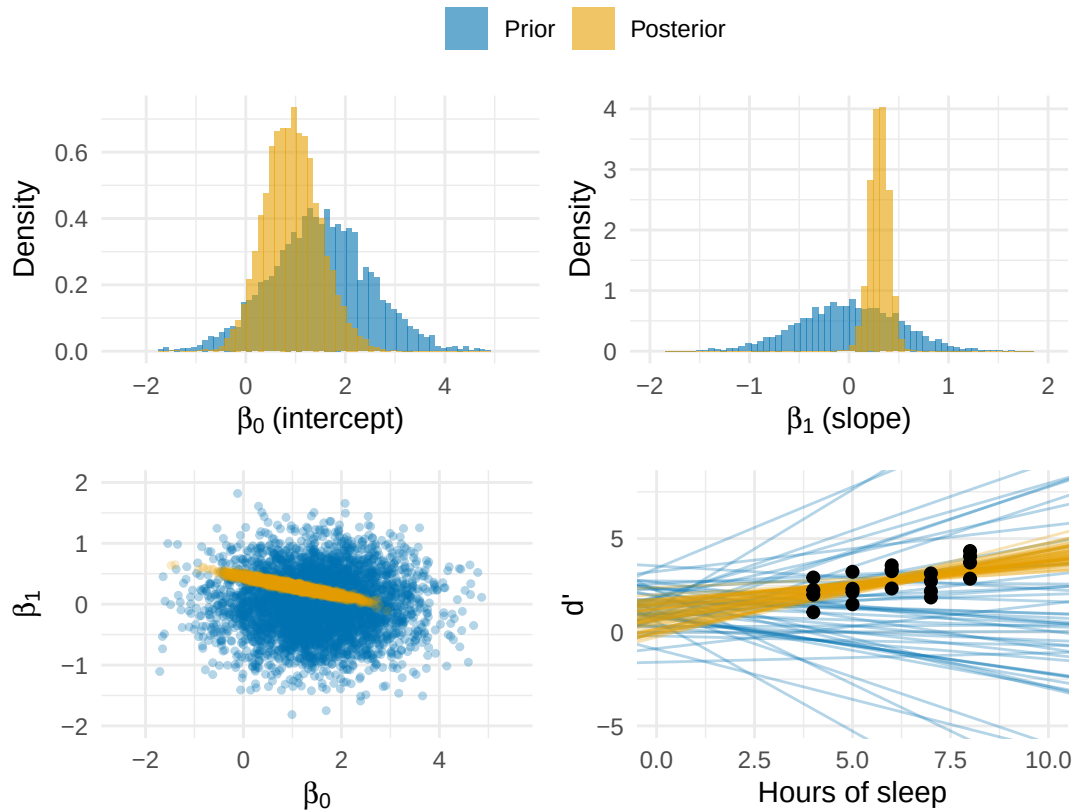


Figure 4: From prior (blue) to posterior (orange). Top: the prior and posterior for β_0 and β_1 . Bottom left: 4000 draws from the joint prior and the joint posterior. Bottom right: 50 lines drawn from the prior and from the posterior, with the 20 observations overlaid (black points).

data (the likelihood). Worlds that were plausible to begin with and that predict the data well survive; the rest fade away. What remains is the **posterior** (Figure 4).

As you can see, the data have narrowed down the plausible worlds considerably. The posteriors for β_0 and β_1 are much narrower than the priors, and of all the lines the prior allowed, only a narrow bundle passing close to the data remains.

Practically

While the above schematic illustrates the idea behind Bayesian model fitting, it is not how model fitting is implemented in practice. Real models usually have many more parameters, and the number of candidate worlds grows exponentially with each one. Even in the simple case above (Figure 4), the posterior is much more concentrated than the prior: if you drew candidates from the prior and weighted each one, you would spend most of your time (and computing power) evaluating candidate worlds that are highly implausible given your data.

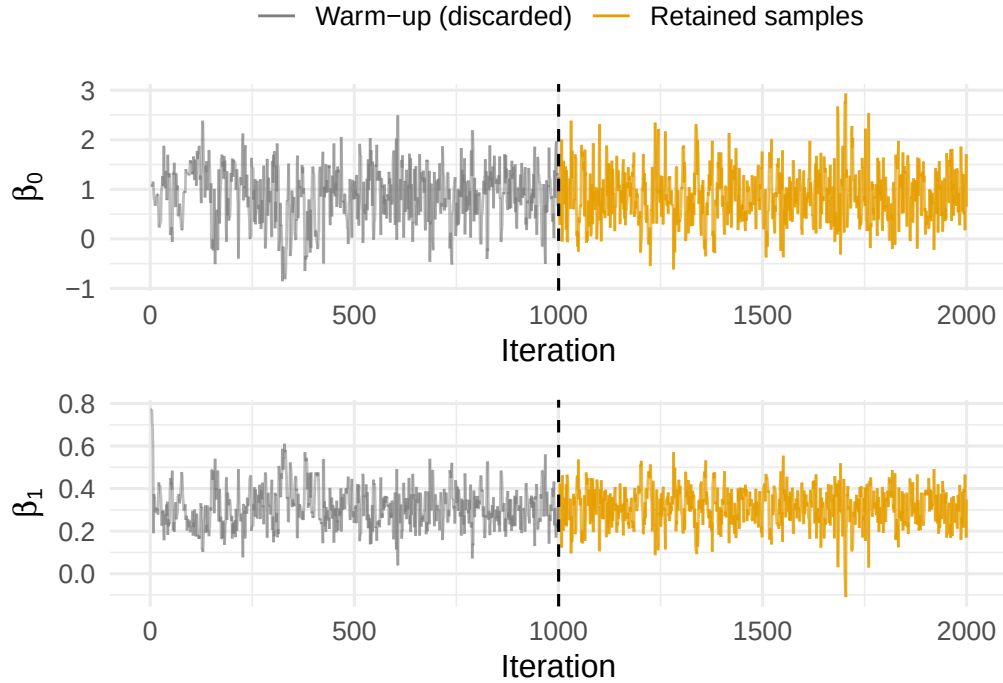


Figure 5: One MCMC chain from our model fit, showing the value of β_0 and β_1 at each step (iteration). The chain starts at a random point; during warm-up (grey) it moves towards the plausible region and tunes its step size. After the dashed line, the retained samples (orange) wander around the posterior; these are the draws used in Figure 4.

Instead, the posterior is obtained using a family of clever computational algorithms called **Markov chain Monte Carlo (MCMC)**.

MCMC methods approximate the posterior without ever computing it in full: they only evaluate prior $\times$ likelihood at the parameter values they visit. Each *chain* works like a walker exploring parameter space: it starts at a random location, and at every step it considers moving to a nearby location. It always moves there if it is more plausible and only sometimes if it is less plausible, and so the chain ends up spending time in each region in proportion to its posterior plausibility¹. The first steps still reflect the chain’s random starting point, so initial samples from this *warm-up* period are discarded; the remaining positions are samples from the posterior, like the orange draws in Figure 4. Figure 5 shows one such chain from our fit.

In practice, several chains are run in parallel, each from a different random starting point. If they all end up exploring the same region, that is reassuring evidence that they have found the posterior, and their retained samples are pooled together. Our fit used four chains of 1000 retained samples each, giving the 4000 posterior draws in Figure 4.

¹Those who are interested to know more may read Chapter 9 of *Statistical Rethinking* by McElreath (2020).

Hypothesis testing with your posterior samples

The posterior samples you get through MCMC sampling represent your updated beliefs about the parameters in your model. They are not an analytical solution, but literally a collection of samples (Table 1). *If* the chains have converged, you can treat these samples as a good approximation of the true posterior, and use them to make inferences about the parameters, and hence about the relative plausibility of the candidate worlds.

Table 1: A few posterior samples from our fit (out of 4000). Each row is one sample from the joint posterior: a combination of parameter values, i.e. one candidate world.

β_0	β_1	σ
0.961	0.321	0.587
0.454	0.365	0.704
0.681	0.321	0.635
0.282	0.424	0.732
1.510	0.225	0.792
1.460	0.220	0.782

Before we look at examples, there is one caveat that applies to everything that follows: **every conclusion is conditional** on the model formula (and its assumptions, such as linearity and normally distributed residuals), the priors, and the data. A statement like “there is a 99.9% probability that the effect is positive” means 99.9% *under these assumptions*.

Because the posterior is just a collection of samples, most questions you might want to ask about your parameters boil down to *counting samples*. For any claim about the parameters, the proportion of samples for which it is true is its **posterior probability**. For example, 3801 of our 4000 samples or 95% have $\beta_1 > 0.15$, so there is a 95% probability that each extra hour of sleep increases d' by more than 0.15 (Figure 6). Another way to summarize the evidence is the **posterior odds** (also called the **evidence ratio**): the number of samples for which the claim is true divided by the number for which it is false. Here, 95% of samples (3801) have $\beta_1 > 0.15$ and 5% (199) have $\beta_1 \leq 0.15$, giving posterior odds of about 19 (95/5). That is, $\beta_1 > 0.15$ is about 19 times more plausible than its alternative $\beta_1 \leq 0.15$. Posterior odds of 1 would mean both are equally likely. As you can see, these two summaries essentially carry the same information: the posterior odds is just the posterior probability that the claim is true divided by the posterior probability that it is false (Figure 6).

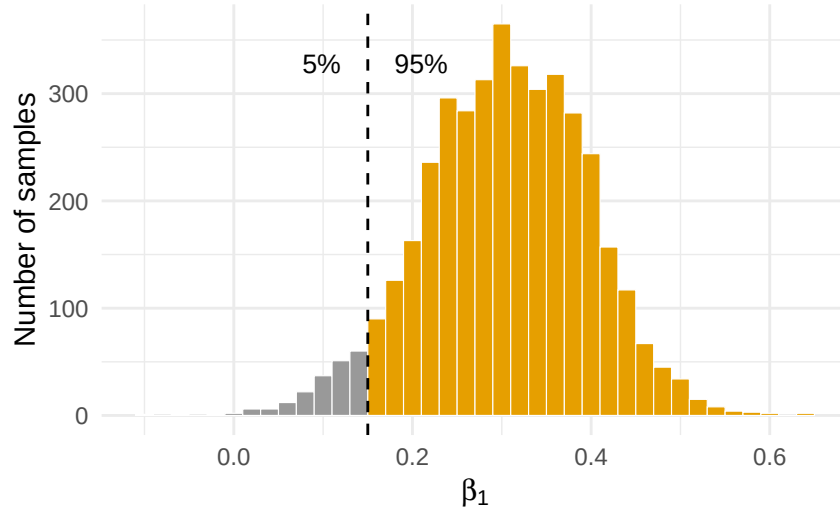


Figure 6: Histogram of the 4000 posterior samples of β_1 . The dashed line marks $\beta_1 = 0.15$. The orange bars are the 3801 samples with $\beta_1 > 0.15$, corresponding to a posterior probability of 95%. The grey bars are the 199 samples with $\beta_1 \leq 0.15$ (posterior probability of 5%). The posterior odds are the ratio of the two counts or posterior probabilities: $3801/199 \approx 95/5 = 19$.

Here are a few examples, all about the effect of sleep, β_1 .

Is the effect positive? We usually have expectations about the direction of an effect. The posterior odds can quantify this by comparing the number of samples with a positive β_1 to the number with a negative β_1 . Here, 3996 of the 4000 samples (99.9%, sometimes reported as the *probability of direction*) are positive, giving posterior odds of 999: a positive effect is about 999 times more plausible than a negative one. So it is very likely that more sleep is associated with higher perceptual sensitivity.

What is the size of the effect? A natural point estimate is the mean of the samples: 0.30. To express the uncertainty around it, find the narrowest range that contains 95% of the samples: here, 0.11 to 0.47. This is the 95% *highest-density interval* (HDI), a type of **credible interval**. Every value inside it is more plausible than any value outside it. In a paper, you might report this as: “Each additional hour of sleep was associated with an increase in d' of 0.30 (95% HDI [0.11, 0.47]).”

Is the effect large enough to matter? You may also want to know whether the effect size is big enough to be meaningful. Suppose an increase of at least 0.2 in d' per hour of sleep is what you would consider *practically significant*: large enough to matter in the real world (not the same as statistically significant). In this example, there is an 88% probability that $\beta_1 > 0.2$.

What do we expect for a particular amount of sleep? The parameters are not always the quantities you care about. For example, you may want to know how well someone is expected

to perform after a short night of 5 hours of sleep, which is $\beta_0 + 5 \times \beta_1$. With posterior samples, you simply compute this for every sample, and the results are samples from the posterior of that quantity. Here, the expected d' after 5 hours of sleep is 2.45 (95% HDI [2.13, 2.80]). The same approach works for any quantity you can compute from the parameters: predictions, differences between conditions, ratios, etc.

Is there an effect of sleep? This is the question null hypothesis significance testing (NHST) asks: is β_1 exactly 0? Among the questions above, this is a rather narrow and uninteresting one: it says nothing about the direction or size of the effect, and the p -value on its own cannot answer those richer questions. For a claim about one exact value like “ $\beta_1 = 0$ ”, counting does not work (no sample is ever *exactly* 0). Instead, we use a **Bayes factor**, which compares how well two hypotheses (or models) predicted the data, here “ $\beta_1 = 0$ ” versus our full model. For a point value hypothesis like this, it can be computed with a mathematical shortcut (the *Savage–Dickey* method): compare the probability density at that value after seeing the data with the probability density at that value before seeing the data (the posterior versus the prior). In our example, the posterior makes $\beta_1 = 0$ less plausible, with a Bayes factor of 0.042. In other words, after seeing the data, “no effect” is about 24 times less plausible than it was before. Unlike a p -value, this is a statement about the parameter itself, not about how surprising the data would be if the null hypothesis was true. Unlike a p -value, Bayes factor can provide evidence *for* the null (a Bayes factor above 1 would mean the data made the null *more* plausible). The frequentist p -value cannot do this; a non-significant result only means the data *failed to reject the null*². Be aware though, that Bayes factors are sensitive to the priors over your parameters, so more considerations are required when assessing evidence for the null.

From numbers to words

Every analysis eventually has to be expressed in words: a conclusion you write in a paper or say in a talk or in a conversation with a colleague. This translation from numerical inference to scientific interpretation is where much of the information in an analysis gets lost.

NHST, as it is usually practiced, makes the translation binary. A p -value below 0.05 becomes “a significant effect”, one above it becomes “no significant effect”. This in practice often gets interpreted as “the effect exists” and “the effect does not exist”. But $p = 0.049$ and $p = 0.051$ carry almost the same evidence, and the 0.05 cutoff is purely a convention (note that this is not an inherent limitation of the frequentist framework, but a result of built up systemic traditions and conventions in science). This issue is compounded by the non-intuitive meaning of the p -value (probability of observing data at least as extreme as your sample if the null hypothesis was true), and by its dependence on the researcher’s intentions (testing and stopping intentions prior to statistical analysis), making it hard to reason about the strength of evidence in a transparent manner.

²An aphorism about this statistical fact goes: “The absence of evidence is not evidence of absence”.

The Bayesian framework, with the posterior, makes more intuitive graded language available. In our example, there is a 99.9% probability that the effect is positive, a 95% probability that it exceeds 0.15, and an 88% probability that it exceeds 0.2. Each supports a statement of a different strength: the effect is almost certainly positive, very likely larger than 0.15, and probably, but not certainly, larger than 0.2. You can report the evidence at the strength it actually has, and readers can judge it by their own standards very intuitively. This clear intuition is enabled by the philosophy of the Bayesian framework, that you maintain beliefs of the relative plausibility of possibilities, and update them with observations in a principled manner. Note that verbal labels for Bayes factors do exist (such as “moderate” or “strong” evidence), but they are rough guides with no rigid conventions in place (yet). Switching to the Bayesian framework is an opportunity to avoid the trap of conventions such as hard cutoffs which would bring back the same black-and-white thinking problem that plagues the frequentist framework.